

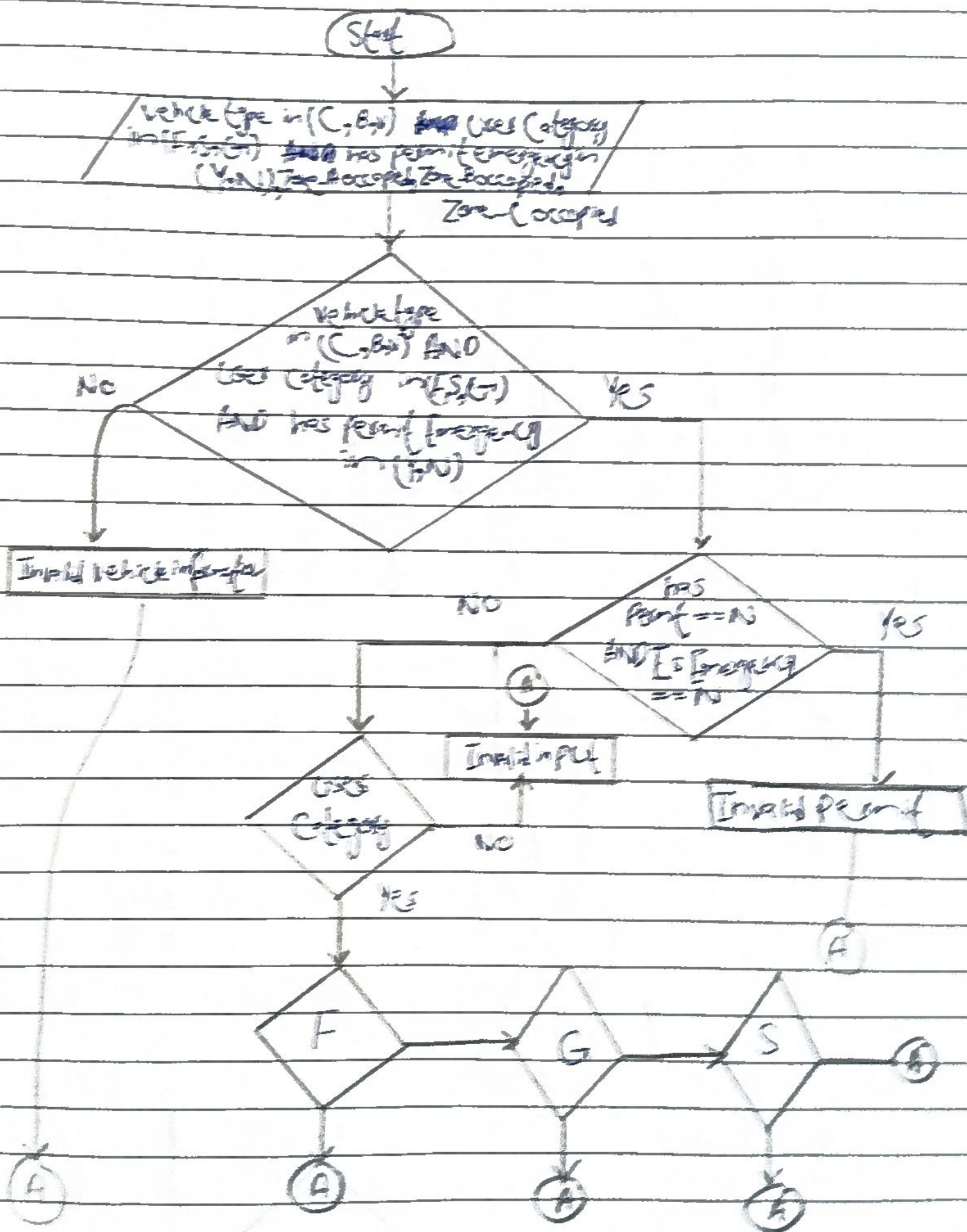
Question 5

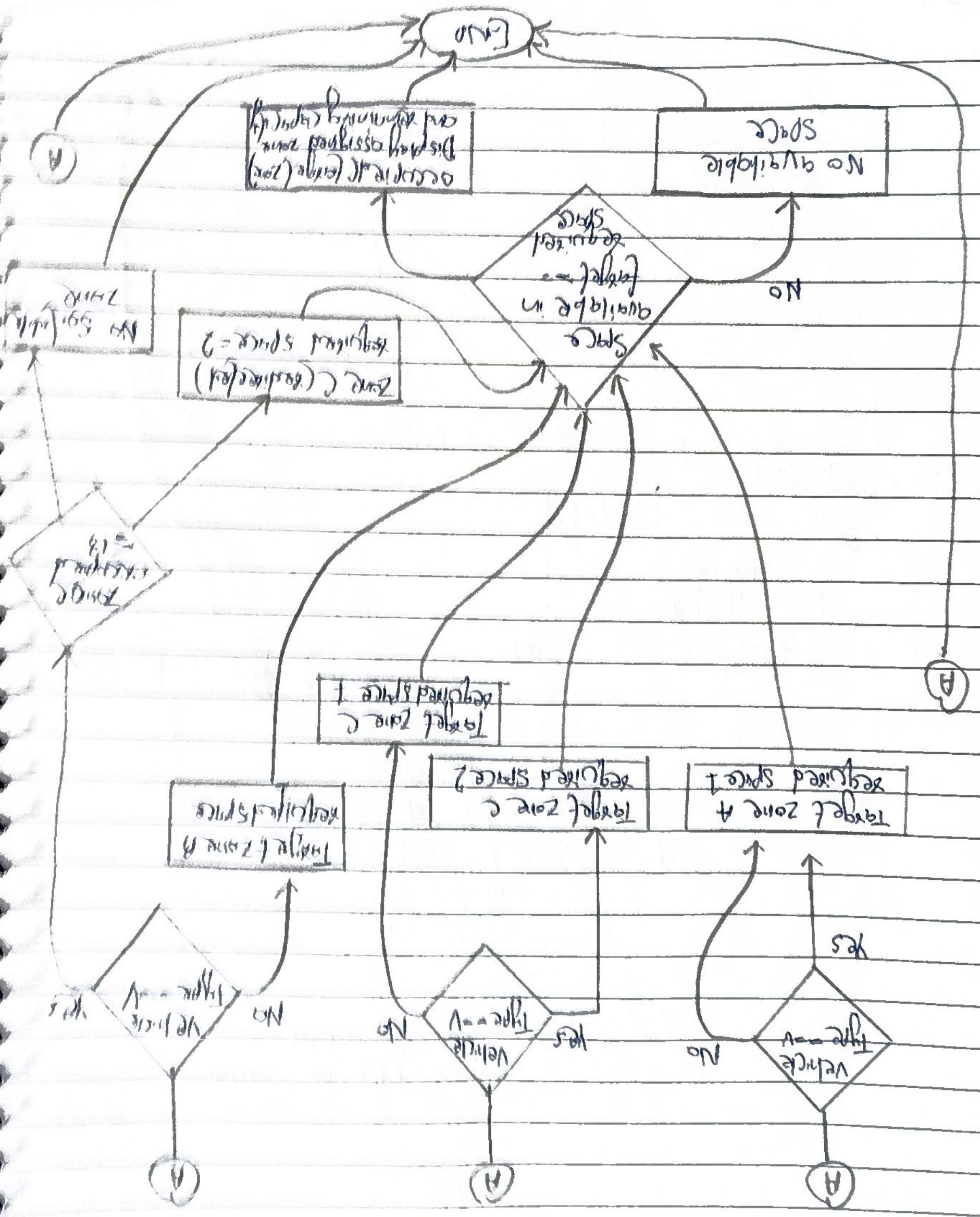
PAC

Input Data	Output	Processing Required
Vehicle Type, User category, has permit, Is Emergency	Assigned zone and remaining capacity (if parked)	Validate vehicle type, User category, has permit, Is emergency
Zone - A occupied, Zone - B occupied, Zone - C occupied (Current occupancy)	Rejection Reason (If Rejected)	Nested decision by User category + Vehicle type to select target zone and required space
Fixed Capacities Zone A=20, Zone B=40, Zone C=15		Check available capacity in target before assigning

IPO

Input	Process	Output
- Vehicle type - User category - has permit - Is Emergency	- Validate Vehicle type, User category, has permit, Is Emergency - If $\Rightarrow$ has permit $\Rightarrow$ N and B - Is Emergency $\Rightarrow$ N $\rightarrow$ Reject (Invalid) - Switch on User category (F/S/G)	- Assigned zone - Remaining Capacity - Rejection reason (if any)
- Zone - A occupied - Zone - B occupied - Zone - C occupied	- nested check on vehicle type $\Rightarrow$ V to pick target zone and required space - Student rejected to zone C if ≥ 2 spaces free, else rejected - Check target zone's available space $\geq$ required space	





Pseudo Code :-

Begin

Constant Capacity A = 20, Capacity B = 40, Capacity C = 15

Input vehicle type, user category, has permit, Is Emergency

Display "Enter current occupied spaces"

Input Zone A _____ occupied, Zone B _____ occupied,

Zone C _____ occupied

IF

{ Vehicle type != "C" AND vehicle type != "B" AND
vehicle type != "V" OR User category != "F" AND
User category != "S" AND User category != "G" }

Display "Rejected : Invalid permit"

ELSE

Suitable zone found = True

Switch (User Category)

Case "F":

target zone = "A"

required space = 1

Case "S":

IF vehicle type = "V"

IF (Capacity C - Zone C - occupied) >= 2

target zone = "C"

required space = 2

ELSE

Suitable zone found = false

END IF

ELSE

Target zone = "B"

required space = 1

END IF

Case "G":

target zone = "C"

IF vehicle type = "V"

required space = 2

ELSE

required space = 1

END IF

END Switch

IF Suitable zone found == false

Display "Rejected: no suitable zone"

ELSE

IF target zone = "A"

available space = Capacity A - Zone A occupied

ELSE

available space = Capacity C - Zone C occupied

END IF

If available space $\geq$ required space

Display "vehicle parked in zone", forget zone

Display "Remaining capacity", (Available space - Required space)

Else

Display "Rejected: No available space"

END IF

END IF

END IF

END